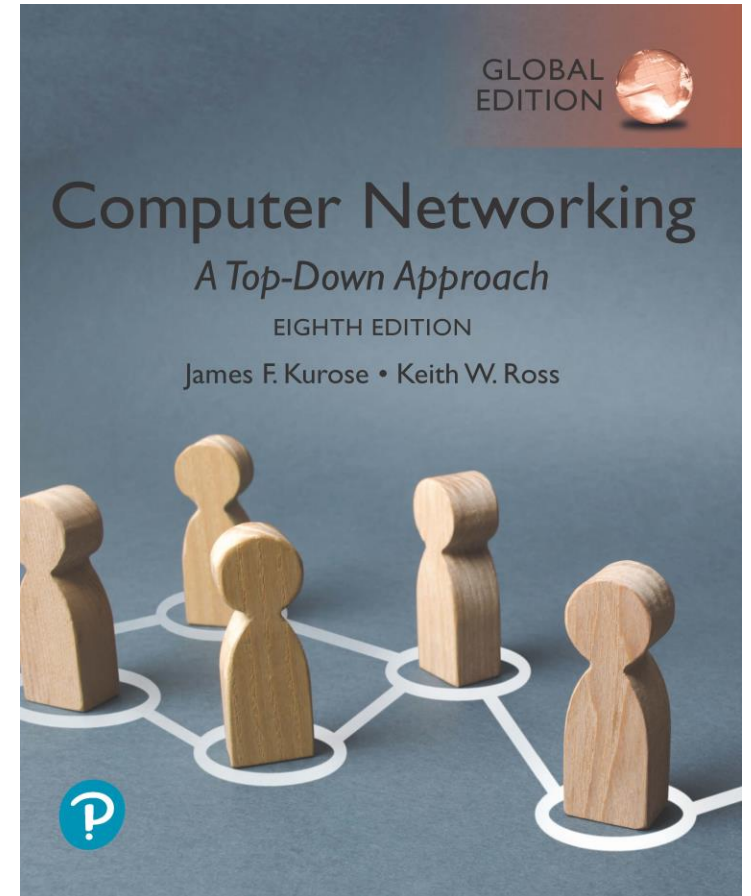


Chapter 8

Security



Computer Networking: A Top-Down Approach

8th edition, Global Edition

Jim Kurose, Keith Ross

Copyright © 2022 Pearson Education Ltd

Chapter 8 outline

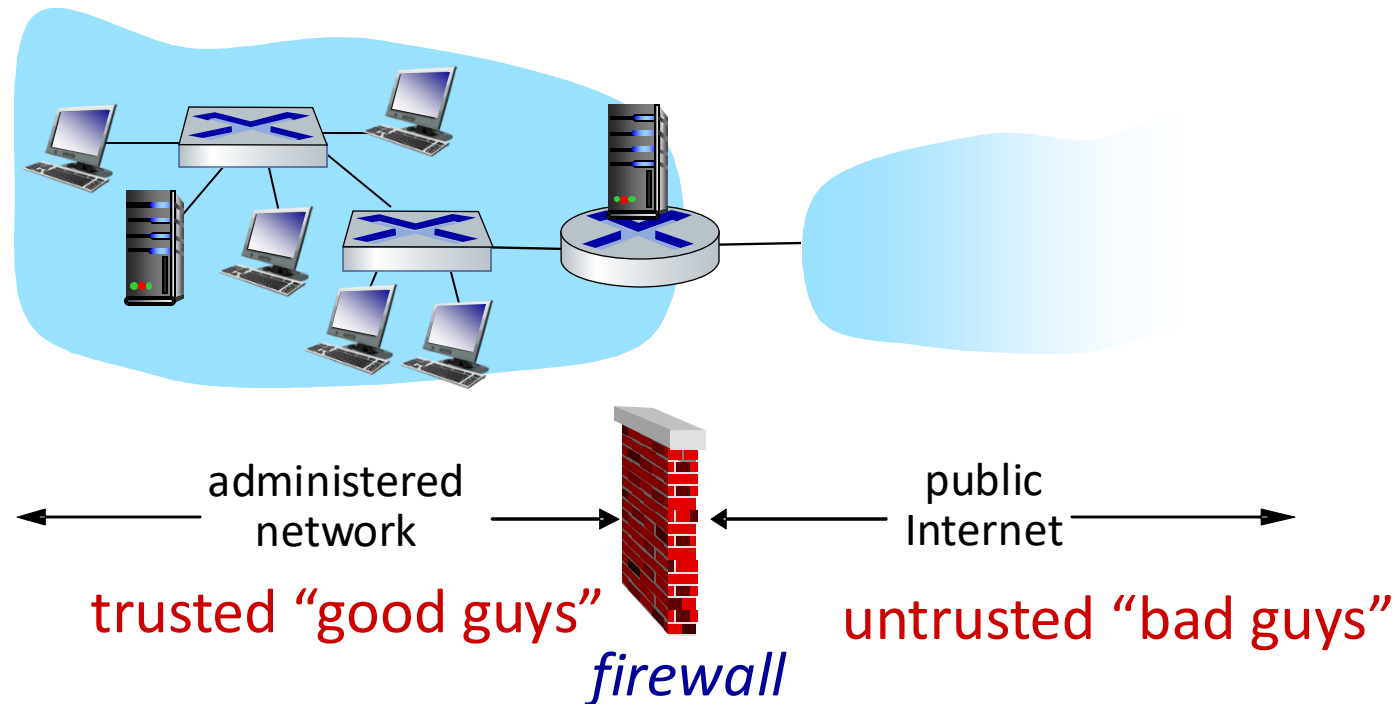
- What is network security?
- Principles of cryptography
- Authentication, message integrity
- Securing e-mail
- Securing TCP connections: TLS
- Network layer security: IPsec
- Security in wireless and mobile networks
- **Operational security: firewalls**



Firewalls

firewall

isolates organization's internal network from larger Internet, allowing some packets to pass, blocking others



Firewalls: why

prevent denial of service attacks:

- SYN flooding: attacker establishes many bogus TCP connections, no resources left for “real” connections

prevent illegal modification/access of internal data

- e.g., attacker replaces CIA’s homepage with something else

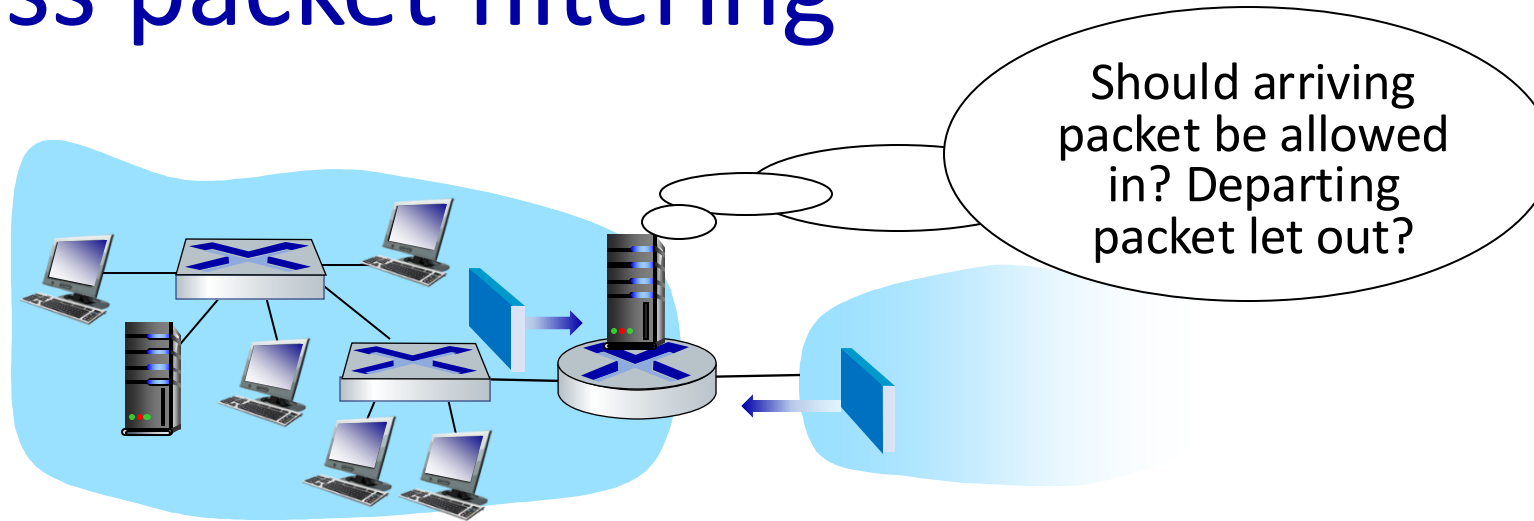
allow only authorized access to inside network

- set of authenticated users/hosts

three types of firewalls:

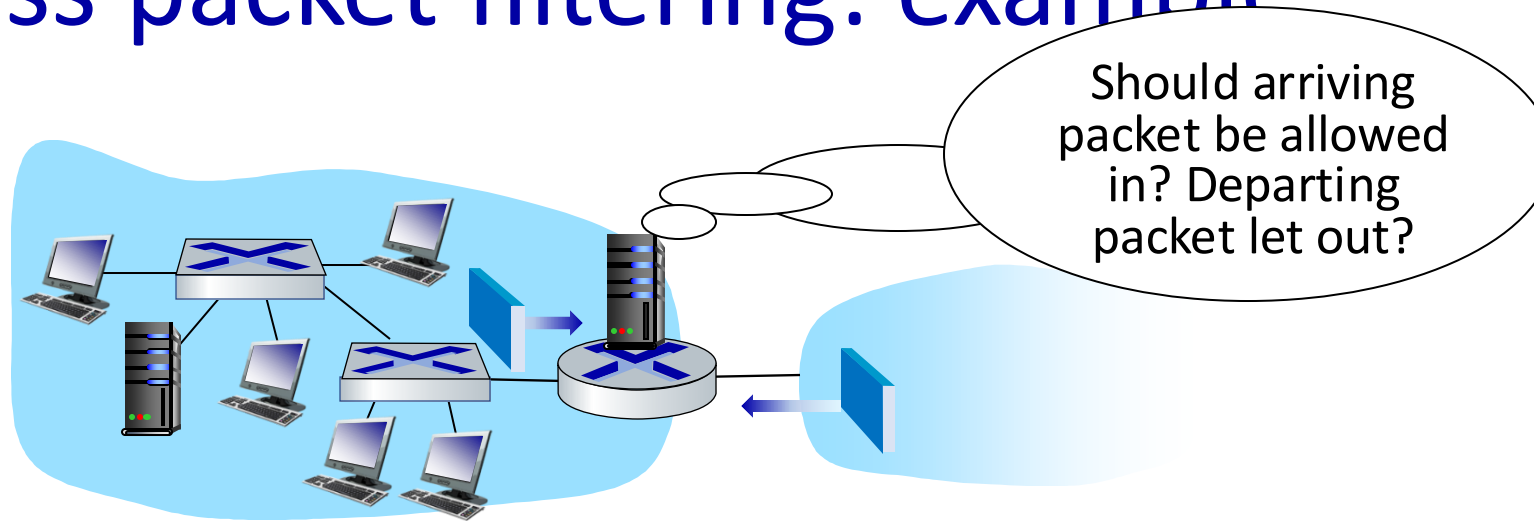
- stateless packet filters
- stateful packet filters
- application gateways

Stateless packet filtering



- internal network connected to Internet via router **firewall**
- filters **packet-by-packet**, decision to forward/drop packet based on:
 - source IP address, destination IP address
 - TCP/UDP source, destination port numbers
 - ICMP message type
 - TCP SYN, ACK bits

Stateless packet filtering: example



- **example 1:** block incoming and outgoing datagrams with IP protocol field = 17 and with either source or dest port = 23
 - **result:** all incoming, outgoing **UDP** flows and telnet connections are blocked
- **example 2:** block inbound TCP segments with ACK=0
 - **result:** prevents external clients from making TCP connections with internal clients, but allows internal clients to connect to outside

Stateless packet filtering: more examples

Policy	Firewall Setting
no outside Web access	drop all outgoing packets to any IP address, port 80
no incoming TCP connections, except those for institution's public Web server only.	drop all incoming TCP SYN packets to any IP except 130.207.244.203, port 80
prevent Web-radios from eating up the available bandwidth.	drop all incoming UDP packets - except DNS and router broadcasts.
prevent your network from being used for a smurf DoS attack.	drop all ICMP packets going to a "broadcast" address (e.g. 130.207.255.255)
prevent your network from being tracerouted	drop all outgoing ICMP TTL expired traffic

Access Control Lists

ACL: table of rules, applied top to bottom to incoming packets: (action, condition) pairs

action	source address	dest address	protocol	source port	dest port	flag bit
allow	222.22/16	100.22/16	TCP	> 1023	80	any
allow	100.22/16	222.22/16	TCP	80	> 1023	ACK
allow	222.22/16	100.22/16	UDP	> 1023	53	---
allow	100.22/16	222.22/16	UDP	53	> 1023	----
deny	all	all	all	all	all	all

Stateful packet filtering

- *stateless packet filter*: heavy handed tool
 - admits packets that “make no sense,” e.g., dest port = 80, ACK bit set, even though no TCP connection established:

action	source address	dest address	protocol	source port	dest port	flag bit
allow	100.22/16	222.22/16	TCP	80	> 1023	ACK

- *stateful packet filter*: track status of every TCP connection
 - track connection setup (SYN), teardown (FIN): determine whether incoming, outgoing packets “makes sense”
 - timeout inactive connections at firewall: no longer admit packets

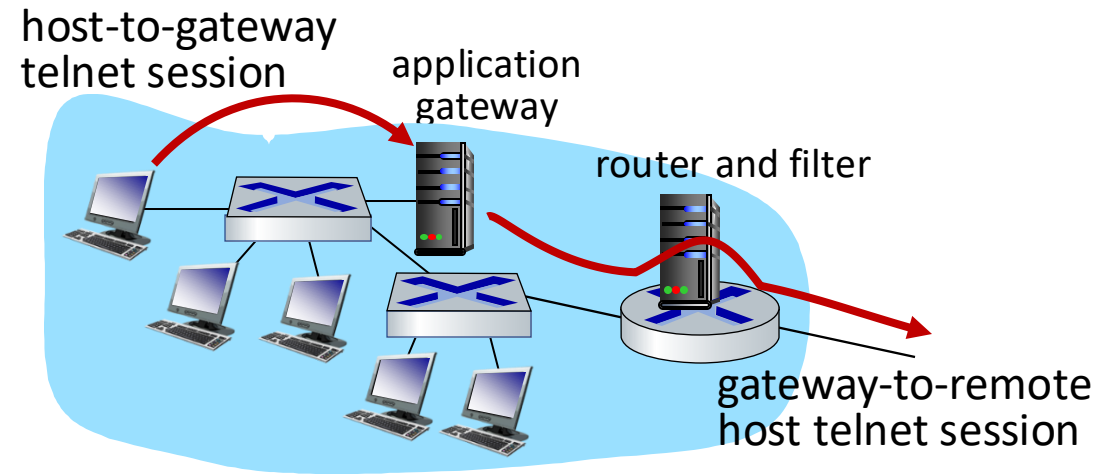
Stateful packet filtering

ACL augmented to indicate need to **check connection state table** before admitting packet

action	source address	dest address	proto	source port	dest port	flag bit	check connection
allow	222.22/16	100.22/16	TCP	> 1023	80	any	
allow	100.22/16	222.22/16	TCP	80	> 1023	ACK	X
allow	222.22/16	100.22/16	UDP	> 1023	53	---	
allow	100.22/16	222.22/16	UDP	53	> 1023	----	X
deny	all	all	all	all	all	all	

Application gateways

- filter packets on **application data** as well as on IP/TCP/UDP fields.
- *example:* allow selected **internal users** to telnet outside



1. require all telnet users to telnet through gateway.
2. for authorized users, gateway sets up telnet connection to dest host
 - gateway relays data between 2 connections
3. router filter blocks all telnet connections not originating from gateway

Limitations of firewalls, gateways

- **IP spoofing:** router can't know if data “really” comes from claimed source
- if multiple apps need special treatment, each has own app. gateway
- client software must know how to contact gateway
 - e.g., must set IP address of proxy in Web browser
- filters often use all or nothing policy for UDP
- *tradeoff:* degree of communication with outside world, level of security
- many highly protected sites still suffer from attacks

Network Security (summary)

basic techniques.....

- cryptography (symmetric and public key)
- message integrity

.... used in many different security scenarios

- secure email
- secure transport (TLS)
- IP sec
- 802.11, 4G/5G

operational security: firewalls

